Exponent Rules

NC.8.EE.1 • Properties of Integer Exponents

Unit 2 • Front

1 Exponent Basics

$$5^3 = 5 \times 5 \times 5 = 125$$

base → 5, exponent → 3

Watch the negative sign!

$(-2)^4 = (-2)(-2)(-2)(-2) = 16$ base is -2

$-2^4 = -(2 \times 2 \times 2 \times 2) = -16$ base is 2

Base: the number being multiplied.
Exponent: how many times the base is a factor.
Read it: "5 to the 3rd power" or "5 cubed"

Why $a^0 = 1$ and $a^{-n} = \frac{1}{a^n}$: follow the pattern. Each step ÷ 2.

$2^3 = 8 \xrightarrow{\div 2} 2^2 = 4 \xrightarrow{\div 2} 2^1 = 2 \xrightarrow{\div 2} \boxed{2^0 = 1} \xrightarrow{\div 2} 2^{-1} = \frac{1}{2^1} = \frac{1}{2} \xrightarrow{\div 2} 2^{-2} = \frac{1}{2^2} = \frac{1}{4} \xrightarrow{\div 2} 2^{-3} = \frac{1}{2^3} = \frac{1}{8}$

2 The Exponent Rules ($a \neq 0$, m and n are integers)

Rule	Pattern	In Words	With Numbers	With Monomials
Product	$a^m \cdot a^n = a^{m+n}$	Multiply → **ADD** exponents	$3^2 \cdot 3^4 = 3^6$	$3x^2 \cdot 4x^5 = 12x^7$
Quotient	$\frac{a^m}{a^n} = a^{m-n}$	Divide → **SUBTRACT** exponents	$\frac{7^5}{7^3} = 7^2 = 49$	$\frac{10y^8}{2y^3} = 5y^5$
Power of a Power	$(a^m)^n = a^{m \cdot n}$	**MULTIPLY** exponents	$(2^3)^2 = 2^6 = 64$	$(m^4)^3 = m^{12}$
Power of a Product	$(ab)^n = a^n b^n$	Exponent to **each** factor	$(2 \cdot 5)^3 = 2^3 \cdot 5^3$	$(2x^3)^2 = 2^2 x^6 = 4x^6$
Power of a Quotient	$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	Exponent to **top and bottom**	$\left(\frac{2}{3}\right)^3 = \frac{8}{27}$	$\left(\frac{x}{3}\right)^2 = \frac{x^2}{9}$
Zero Exponent	$a^0 = 1$	Always 1	$9^0 = 1$	$(7x)^0 = 1$ $5x^0 = 5 \cdot 1 = 5$
Negative Exponent	$a^{-n} = \frac{1}{a^n}$	**Flip** it. Exponent turns positive.	$4^{-2} = \frac{1}{4^2} = \frac{1}{16}$	$x^{-3} = \frac{1}{x^3}$ $4x^{-2} = \frac{4}{x^2}$

Monomials: work the **numbers** (coefficients) first, then use the rule on the **variables**. Only the base with the exponent changes: in $5x^0$, only x becomes 1.

3 Put It Together

Order of steps

1. Powers of powers **first**
2. Combine same bases
3. Make negative exponents positive
4. Evaluate

Power of power: $\frac{(3^2)^4 \cdot 3^{-5}}{3} = \frac{3^8 \cdot 3^{-5}}{3^1}$

Add: $8 + (-5)$ $= \frac{3^3}{3^1}$

Subtract: $3 - 1$ $= 3^2$

$= \boxed{9}$

Power of product: $\frac{(2x^3)^2 \cdot x^{-4}}{x} = \frac{4x^6 \cdot x^{-4}}{x^1}$

Add: $6 + (-4)$ $= \frac{4x^2}{x^1}$

Subtract: $2 - 1$ $= \boxed{4x}$

Common mistakes $2^3 \cdot 2^4 \neq 4^7$ **Keep the base:** 2^7 $5^2 \neq 10$ **It's** $5 \times 5 = 25$ $4^{-2} \neq -16$ **It's** $\frac{1}{16}$ $3^0 \neq 0$ **It's** 1

Scientific Notation

NC.8.EE.3 • NC.8.EE.4

4 What Is Scientific Notation?

$$a \times 10^{n}$$

$1 \le a < 10$ — n = places moved

YES ✓ 3.2×10^5 7×10^{-3}

NO × 32×10^4 (32 is too big)

NO × 0.32×10^6 (0.32 is too small)

10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}
1,000	100	10	1	0.1	0.01	0.001

Calculator: 3.2E5 means 3.2×10^5

5 Convert Between Forms

Standard Form → Scientific Notation

1. Put the decimal after the **first non-zero digit**.
2. Count how many places it moved.
3. 10 or more → **positive** exponent.
   Less than 1 → **negative** exponent.

45,000,000 moved 7 left = $\boxed{4.5 \times 10^7}$

0.00062 moved 4 right = $\boxed{6.2 \times 10^{-4}}$

Scientific Notation → Standard Form

Positive exponent → decimal **right** (bigger).
Negative exponent → decimal **left** (smaller).
Fill empty places with **zeros**.

3.08×10^5 right 5 = $\boxed{308{,}000}$

7.1×10^{-3} left 3 = $\boxed{0.0071}$

6 Estimate and Compare

Estimate: round to **one** digit.

$4{,}890{,}000 \approx 5 \times 10^6$

$0.000312 \approx 3 \times 10^{-4}$

Compare:

1. Bigger exponent wins.
   $2 \times 10^8 > 9 \times 10^7$
2. Same exponent? Compare a.
   $6.1 \times 10^{-3} > 2.5 \times 10^{-3}$

How many times as much? Divide!

Whale: 2×10^5 kg Mouse: 2×10^{-2} kg

$$\frac{2 \times 10^5}{2 \times 10^{-2}} = 1 \times 10^{5-(-2)} = 10^7$$

Whale is **10,000,000 times** as heavy.

7 Multiply and Divide

Multiply: $(2.5 \times 10^3)(3 \times 10^4)$

1. Multiply a's

$$\begin{array}{r} 2.5 \\ \times \quad 3 \\ \hline 7.5 \end{array}$$

2. **ADD** exponents: $3 + 4 = 7$
3. Is a between 1 and 10? 7.5 Yes! ✓

$\boxed{7.5 \times 10^7}$

Divide: $\dfrac{8.4 \times 10^6}{2 \times 10^2}$

1. Divide a's

$$\begin{array}{r} 4.2 \\ 2\overline{)\,8.4} \\ -\,8 \\ \hline 4 \\ -\,4 \\ \hline 0 \end{array}$$

2. **SUBTRACT** exponents: $6 - 2 = 4$
3. Is a between 1 and 10? 4.2 Yes! ✓

$\boxed{4.2 \times 10^4}$

Fix it if a is not between 1 and 10:

$12 \times 10^7 = \boxed{1.2 \times 10^8}$ decimal **left** 1, exponent **up** 1 $0.5 \times 10^4 = \boxed{5 \times 10^3}$ decimal **right** 1, exponent **down** 1

Mixed forms? Change both to scientific notation first: $3{,}000 \times (2 \times 10^5) = (3 \times 10^3)(2 \times 10^5) = 6 \times 10^8$

8 Word Problem Clue Words

MULTIPLY × (exponents **ADD**)

• "**times as much** as" • "**of**" • "**product**"
• "**each**" / "**per**" . . . "how many **in all**?" / "**total**"

Ex: 3×10^4 cells. **Each** is 2×10^{-5} m long.
Total length? $(3 \times 10^4)(2 \times 10^{-5}) = 6 \times 10^{-1}$ m

DIVIDE ÷ (exponents **SUBTRACT**)

• "**how many times** larger / greater?" • "**quotient**"
• "**per** / **each**" (find one) • "**shared equally**"

Ex: $\$8 \times 10^6$ **shared equally** by 2×10^3 people.

Each gets? $\dfrac{8 \times 10^6}{2 \times 10^3} = \$4 \times 10^3$